

```
In [1]: #pip install --upgrade seaborn
```

```
In [2]: import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
```

```
In [3]: import seaborn as sns
```

```
In [4]: sns.get_dataset_names()
```

```
Out[4]: ['anagrams',
         'anscombe',
         'attention',
         'brain_networks',
         'car_crashes',
         'diamonds',
         'dots',
         'dowjones',
         'exercise',
         'flights',
         'fmri',
         'geyser',
         'glue',
         'healthexp',
         'iris',
         'mpg',
         'penguins',
         'planets',
         'seaice',
         'taxi',
         'tips',
         'titanic']
```

```
In [5]: tips = sns.load_dataset("tips")
tips.head()
```

```
Out[5]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
In [6]: tips
```

```
Out[6]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

```
In [7]: import os
os.getcwd()
```

```
Out[7]: 'C:\\Users\\Admin'
```

```
In [8]: import matplotlib.pyplot as plt
```

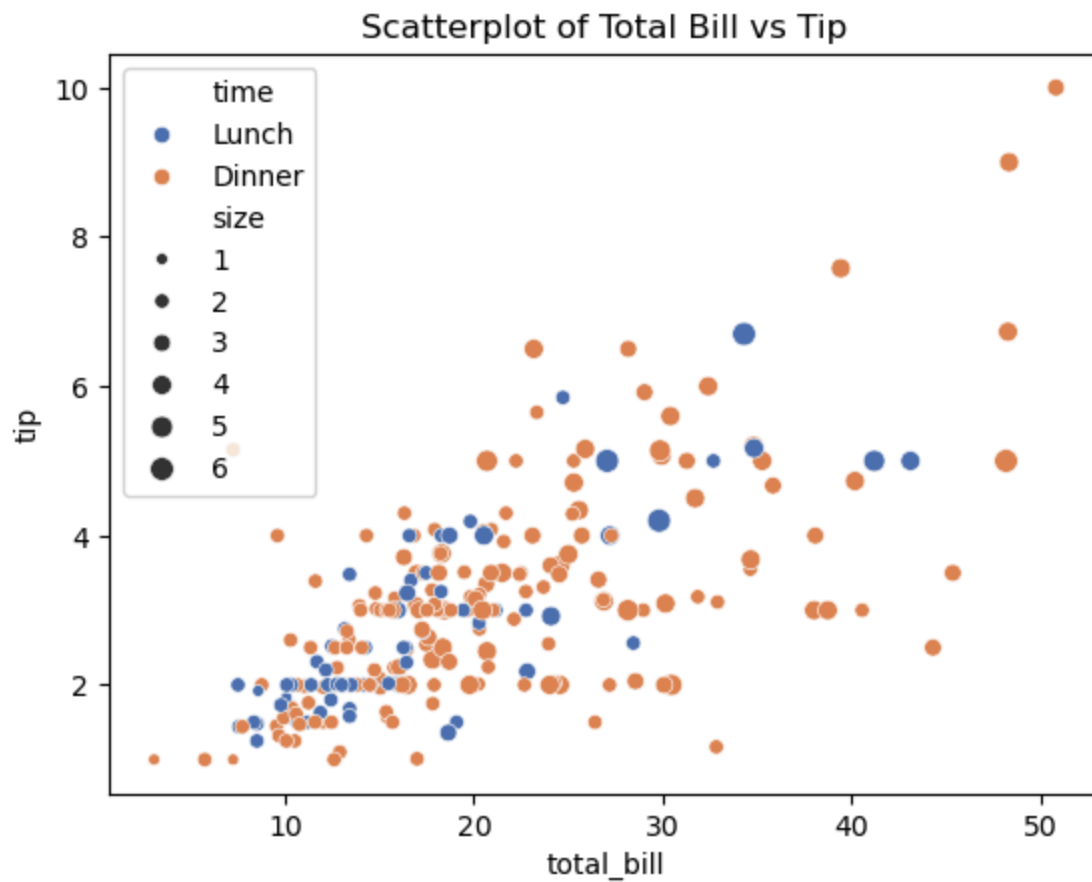
```
In [9]: plt.figure(figsize=(8, 6))
```

```
Out[9]: <Figure size 800x600 with 0 Axes>
```

```
<Figure size 800x600 with 0 Axes>
```

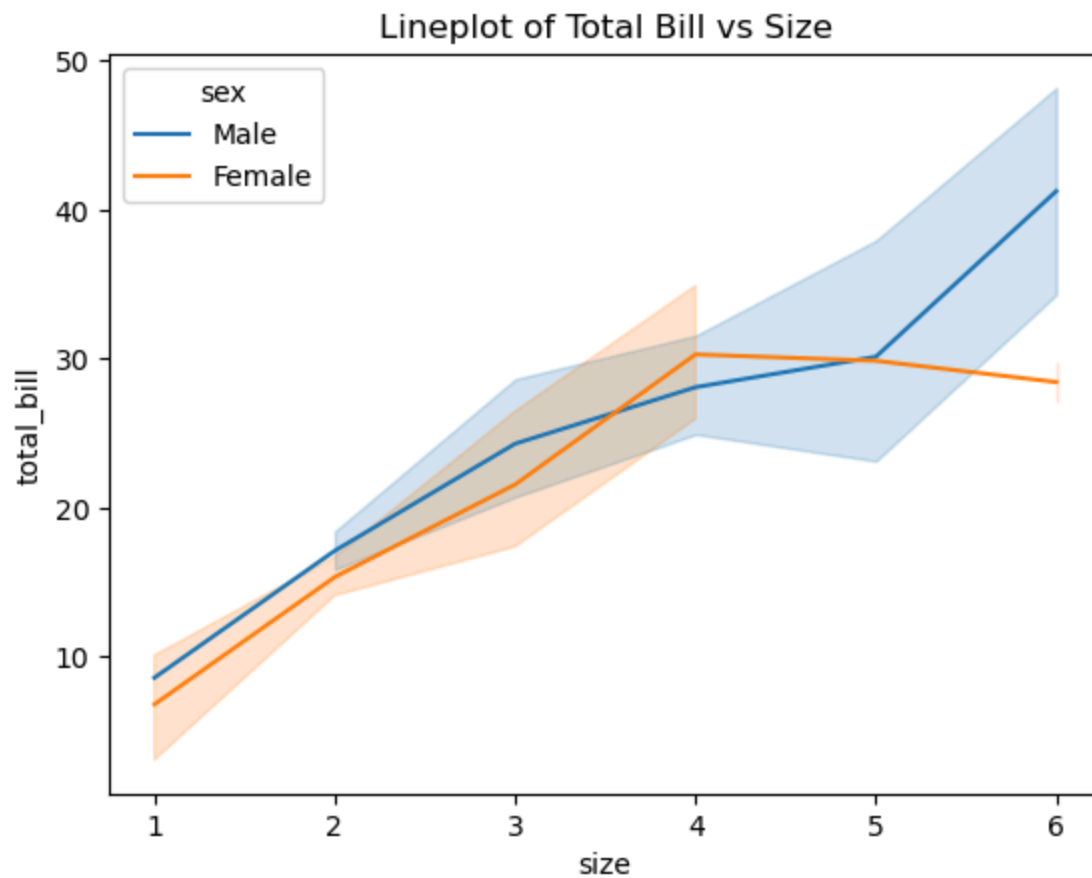
```
In [10]: # 1. scatter plot

sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", size="size", palette
plt.title("Scatterplot of Total Bill vs Tip")
plt.show()
```

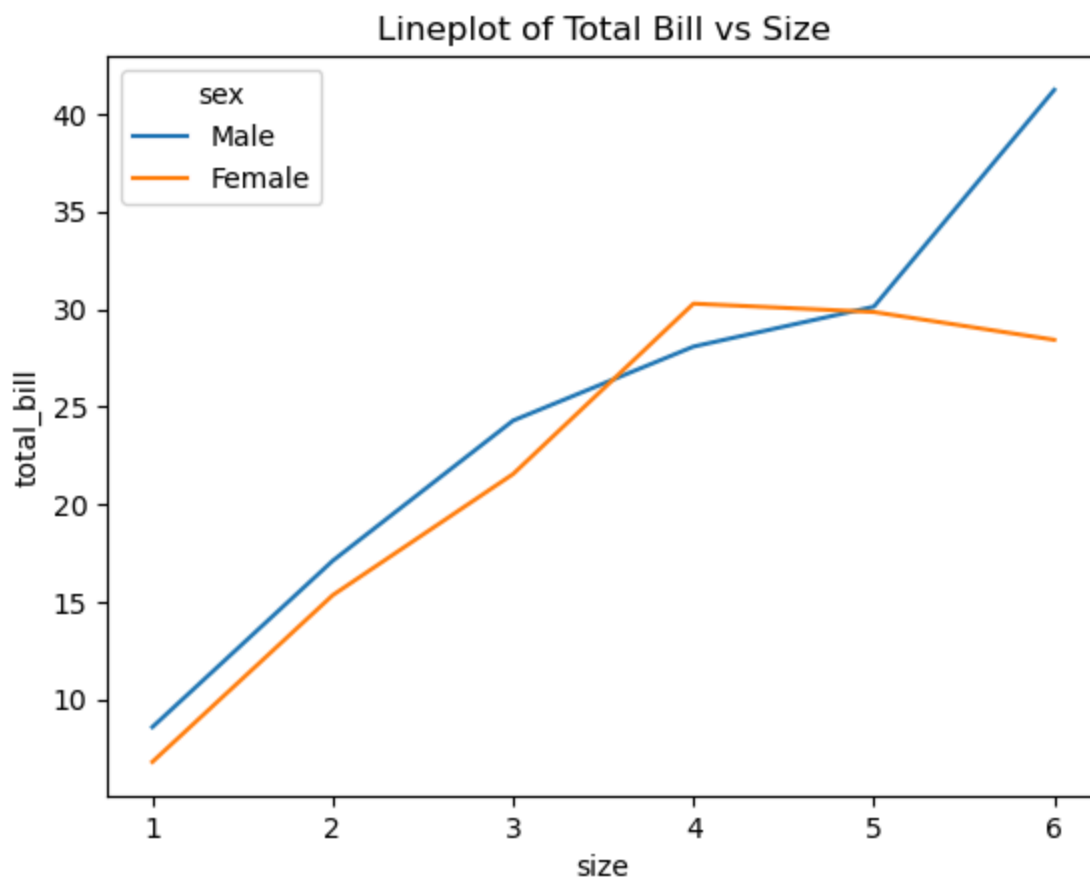


In [11]: `# line plot`

```
sns.lineplot(data=tips, x='size', y='total_bill', hue='sex', markers='o')  
plt.title("Lineplot of Total Bill vs Size")  
plt.show()
```



```
In [12]: sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',ci=None, markers='o')  
plt.title("Lineplot of Total Bill vs Size")  
plt.show()
```

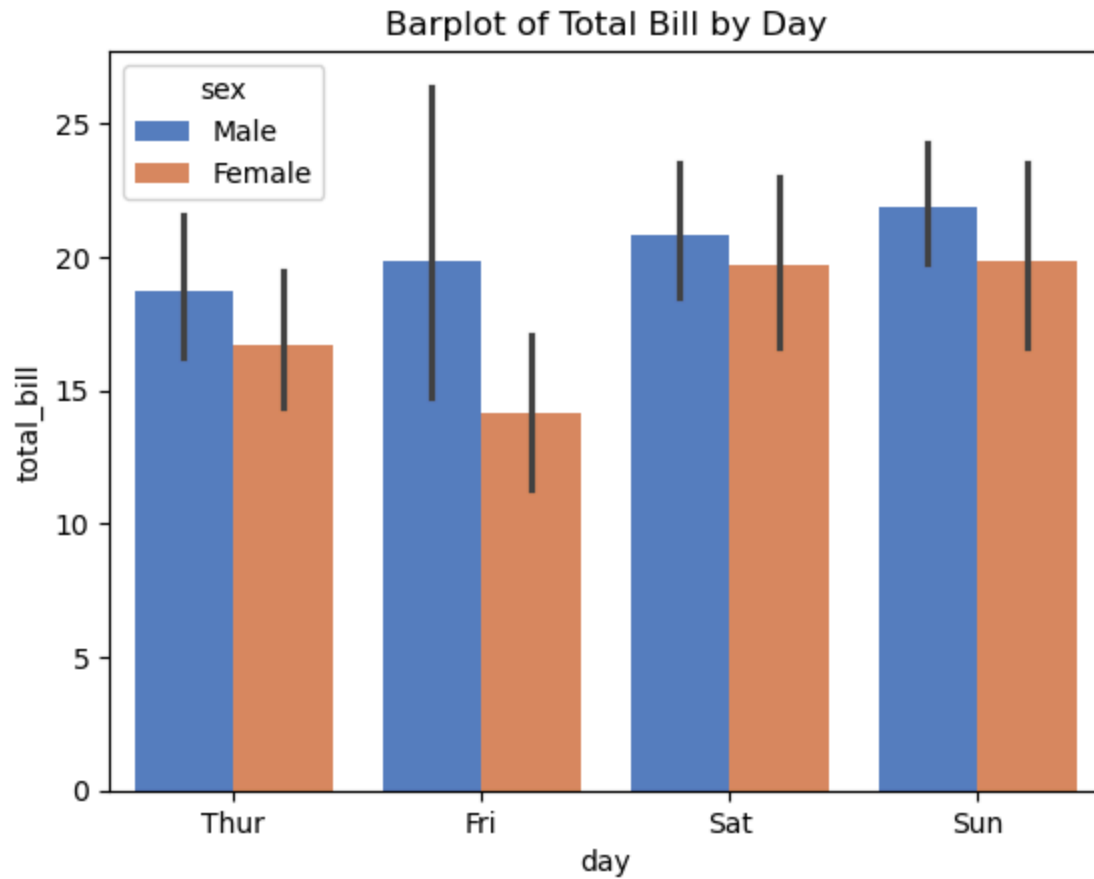


```
In [13]: tips.columns
```

```
Out[13]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

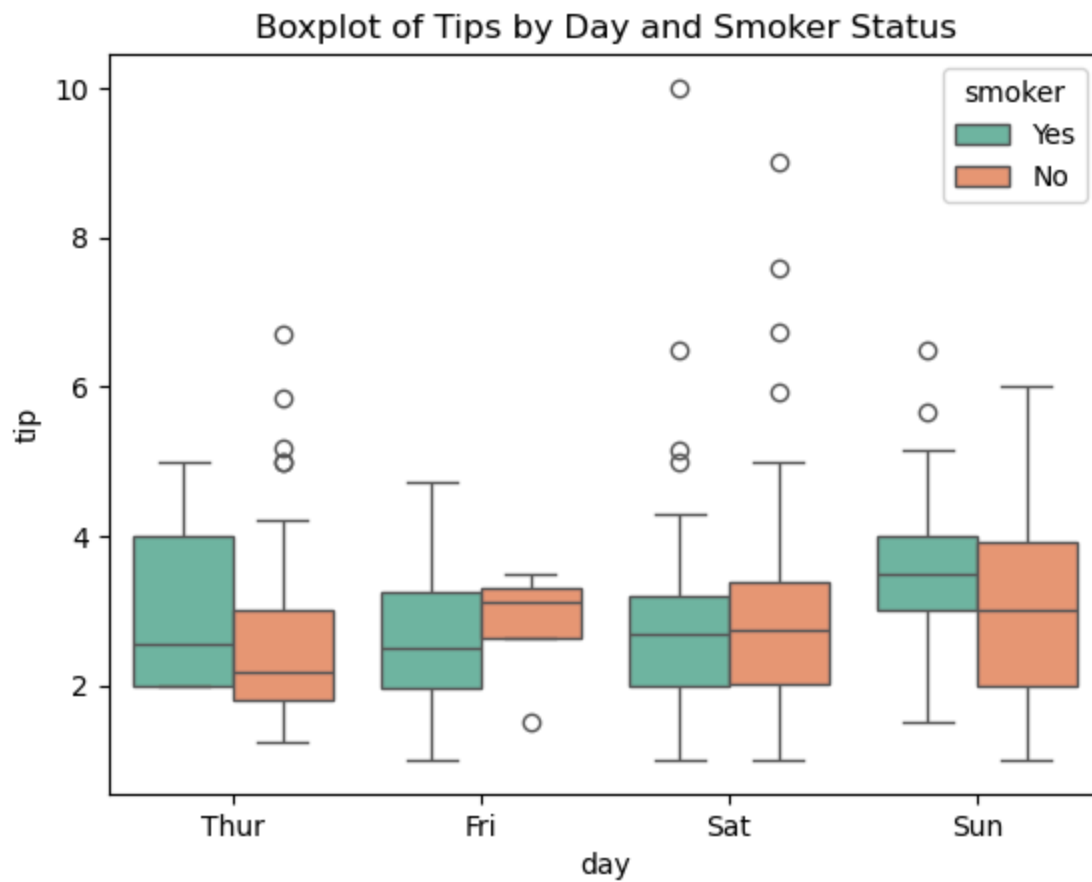
```
In [14]: #3. bar plot
```

```
sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted')  
plt.title("Barplot of Total Bill by Day")  
plt.show()
```



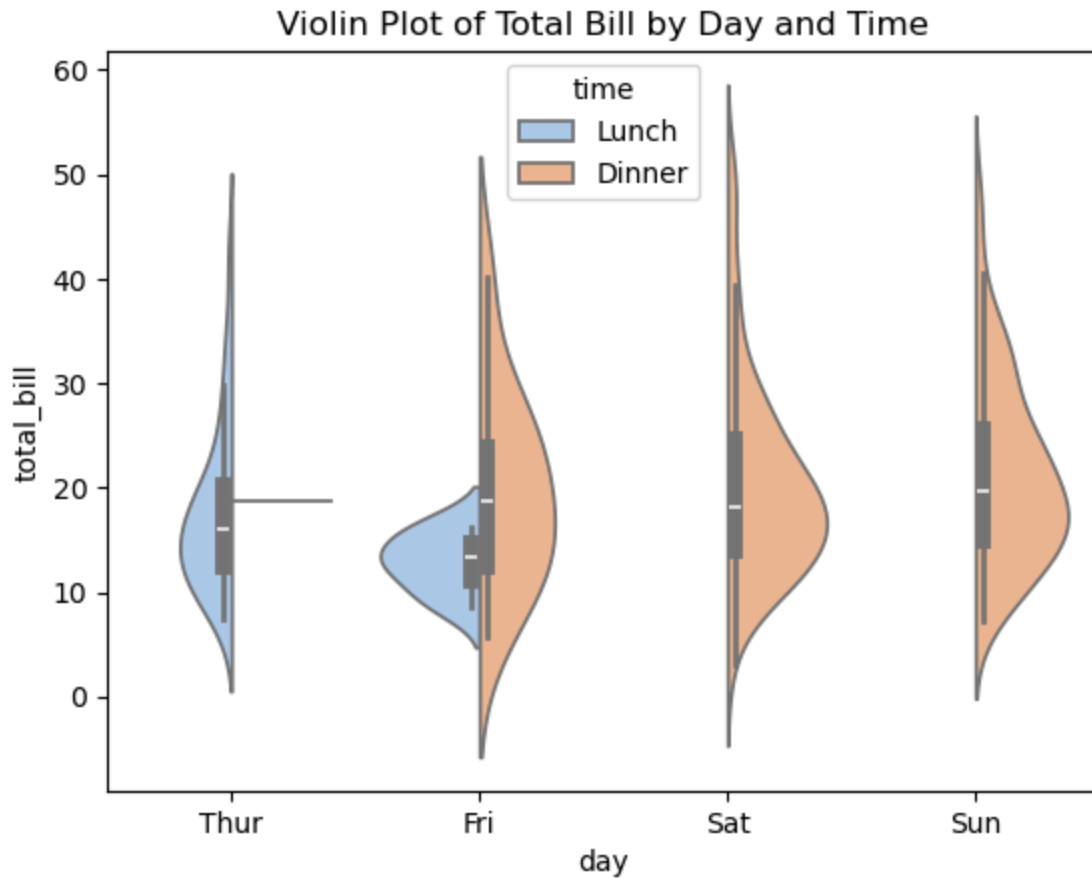
In [15]: # 4. Boxplot

```
sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2')  
plt.title("Boxplot of Tips by Day and Smoker Status")  
plt.show()
```



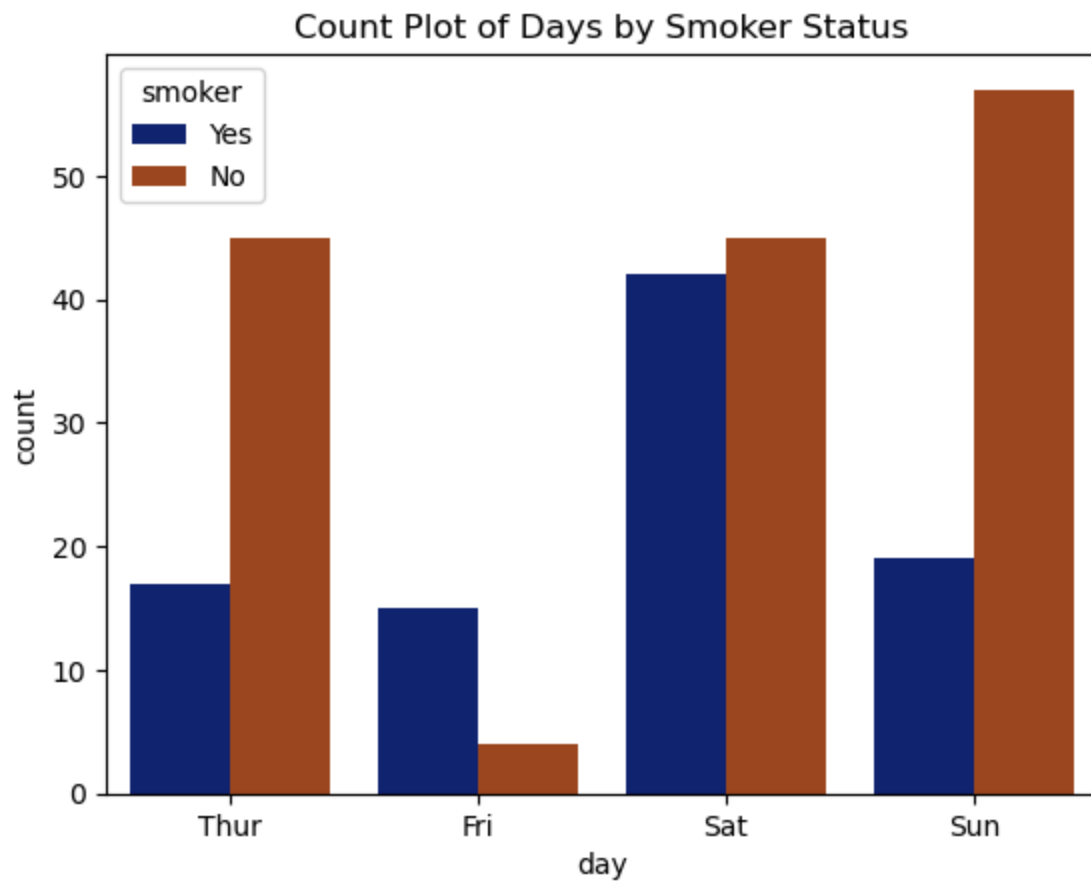
In [16]: # 5. violin plot

```
sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True, palette=
plt.title("Violin Plot of Total Bill by Day and Time")
plt.show()
```



In [17]: #6. count plot

```
sns.countplot(data=tips, x='day', hue='smoker', palette='dark')  
plt.title("Count Plot of Days by Smoker Status")  
plt.show()
```

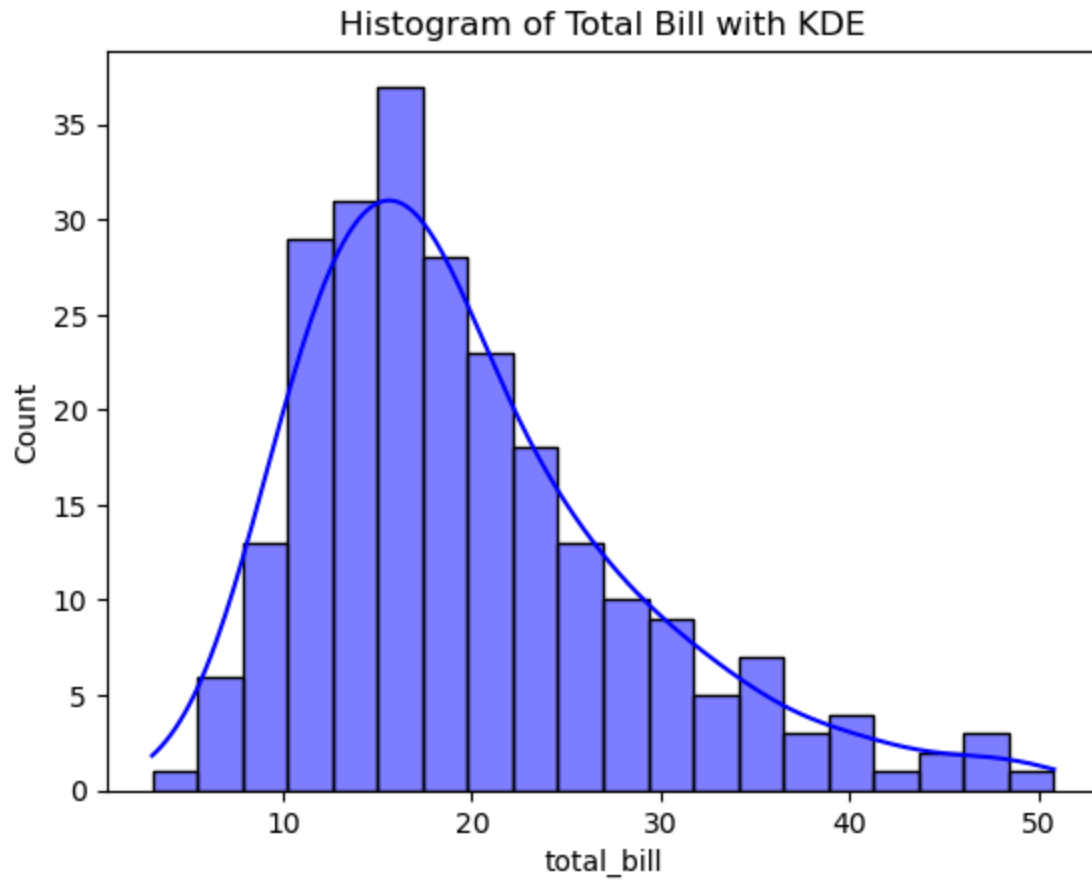
In [18]: #7. regression plot

```
sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50}, line_kws={'co
plt.title("Regression Plot of Total Bill vs Tip")
plt.show()
```



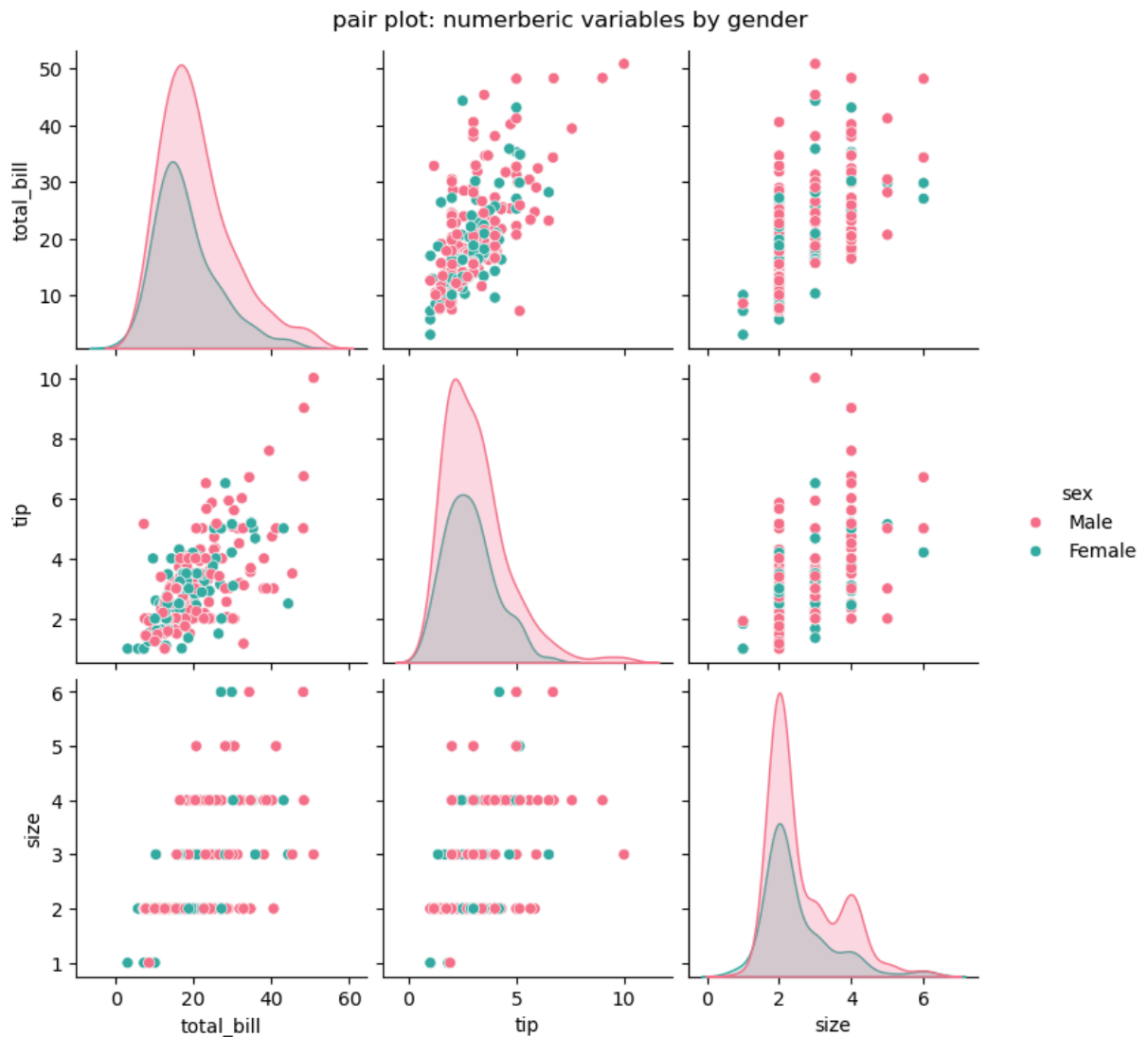
In [19]: *# 8. Histogram of total bill with KDE*

```
sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue')  
plt.title("Histogram of Total Bill with KDE")  
plt.show()
```



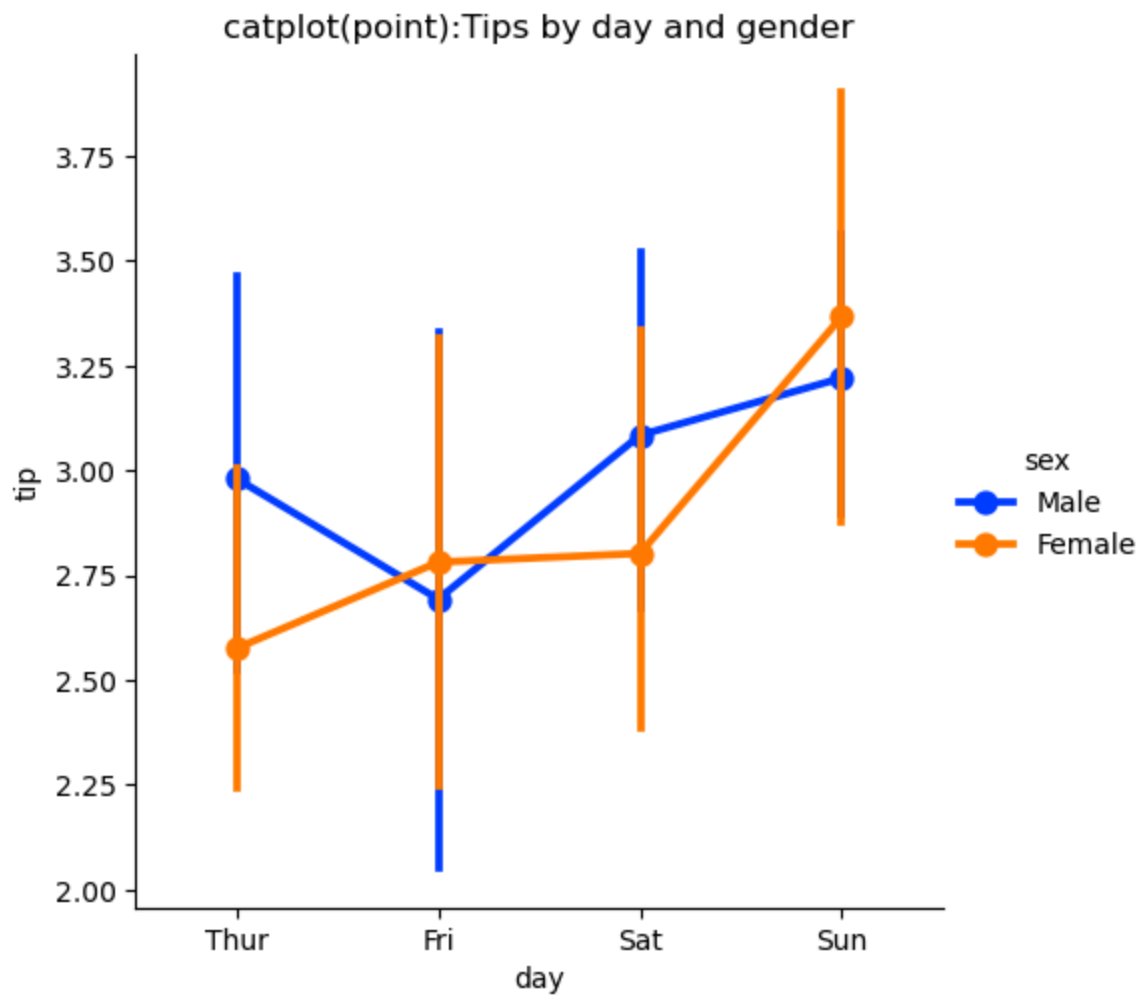
In [20]: #9. pairplot

```
sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"], palette='husl')  
plt.suptitle("pair plot: numeric variables by gender", y=1.02)  
plt.show()
```



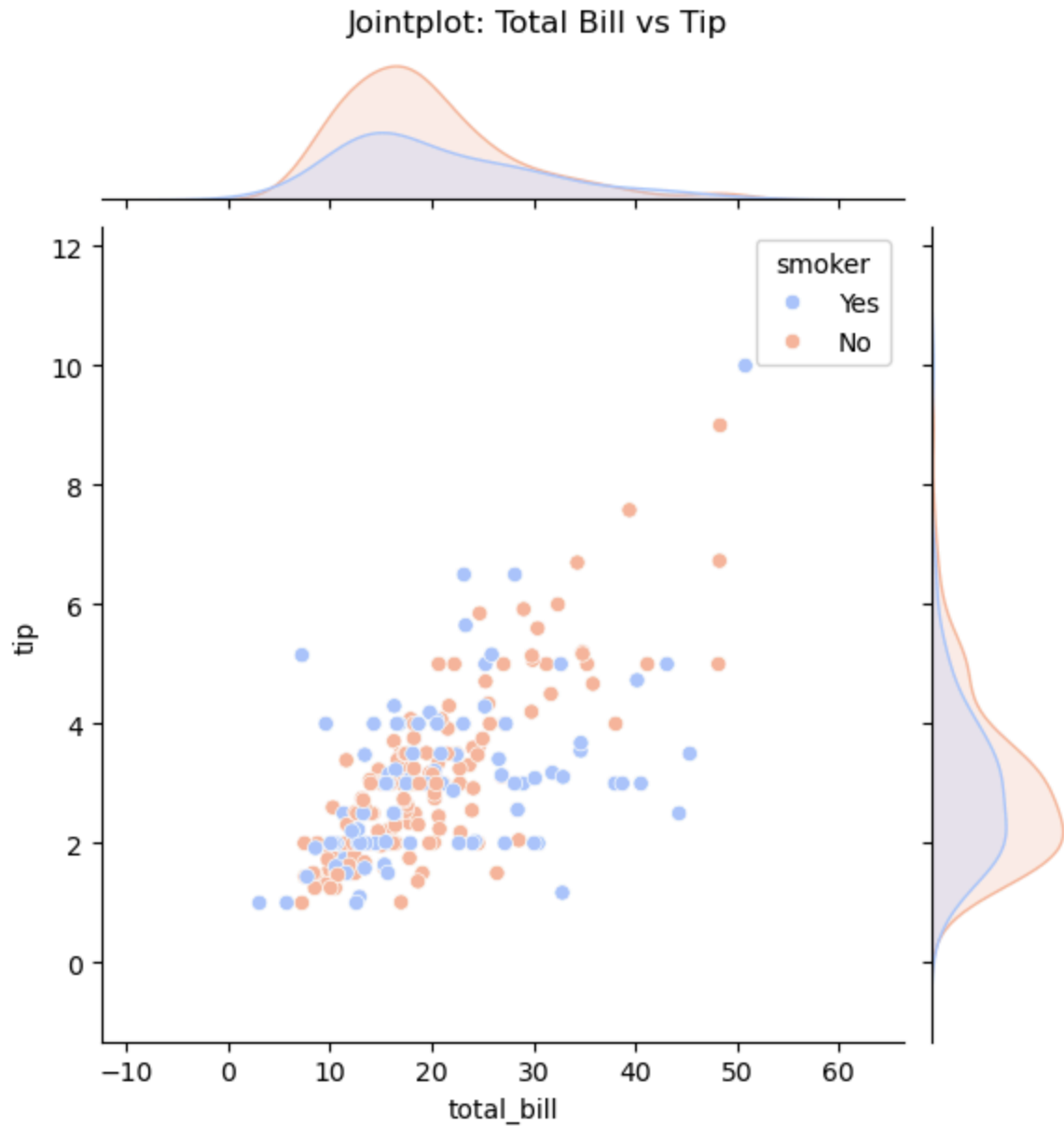
```
In [21]: # 10 catplot

sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point', palette='bright')
plt.title("catplot(point):Tips by day and gender")
plt.show()
```



In [22]: # 11. jointplot

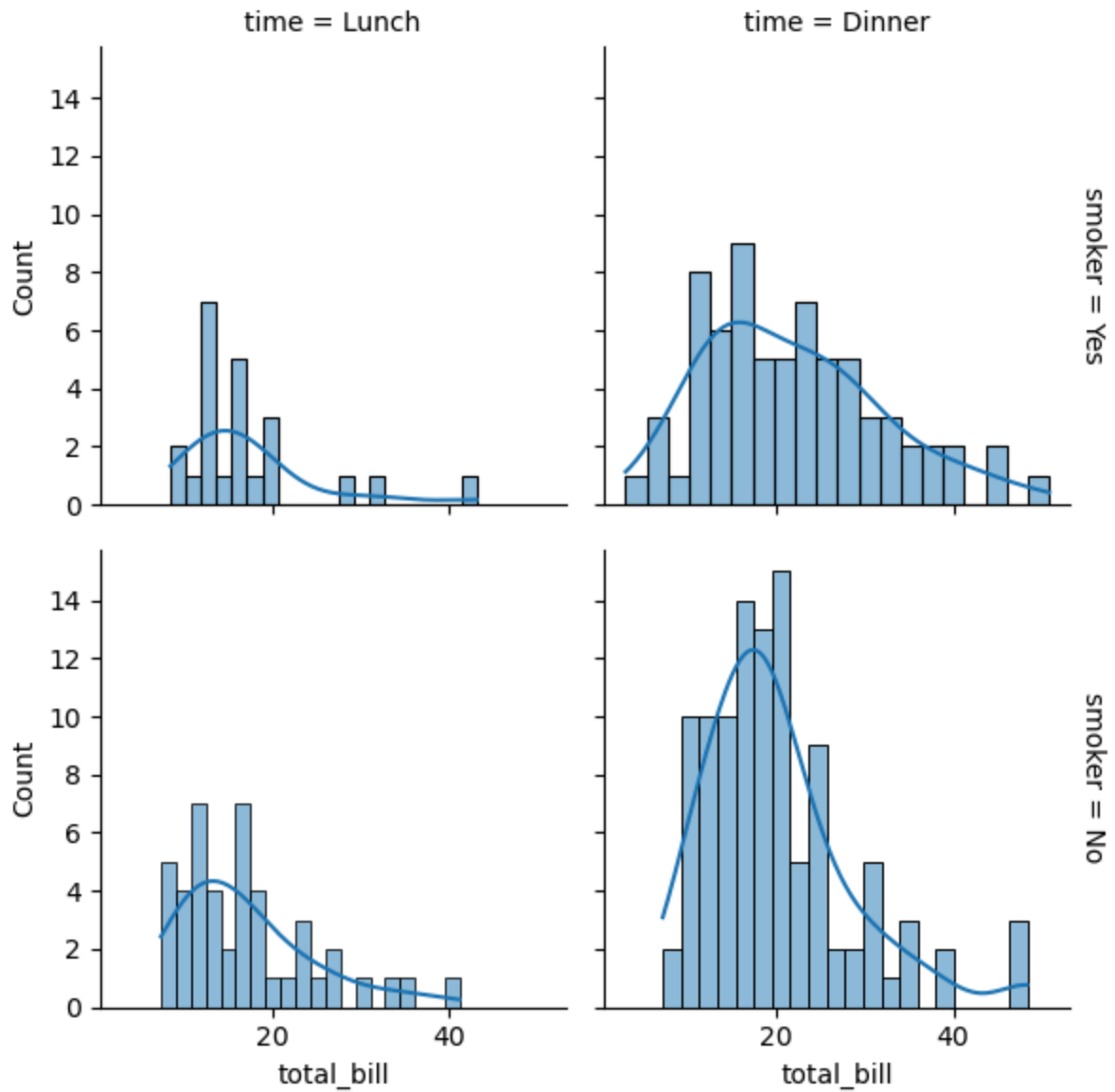
```
sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter', hue='smoker', col=1, row=1,
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)
plt.show()
```



```
In [23]: # Facetgrid
```

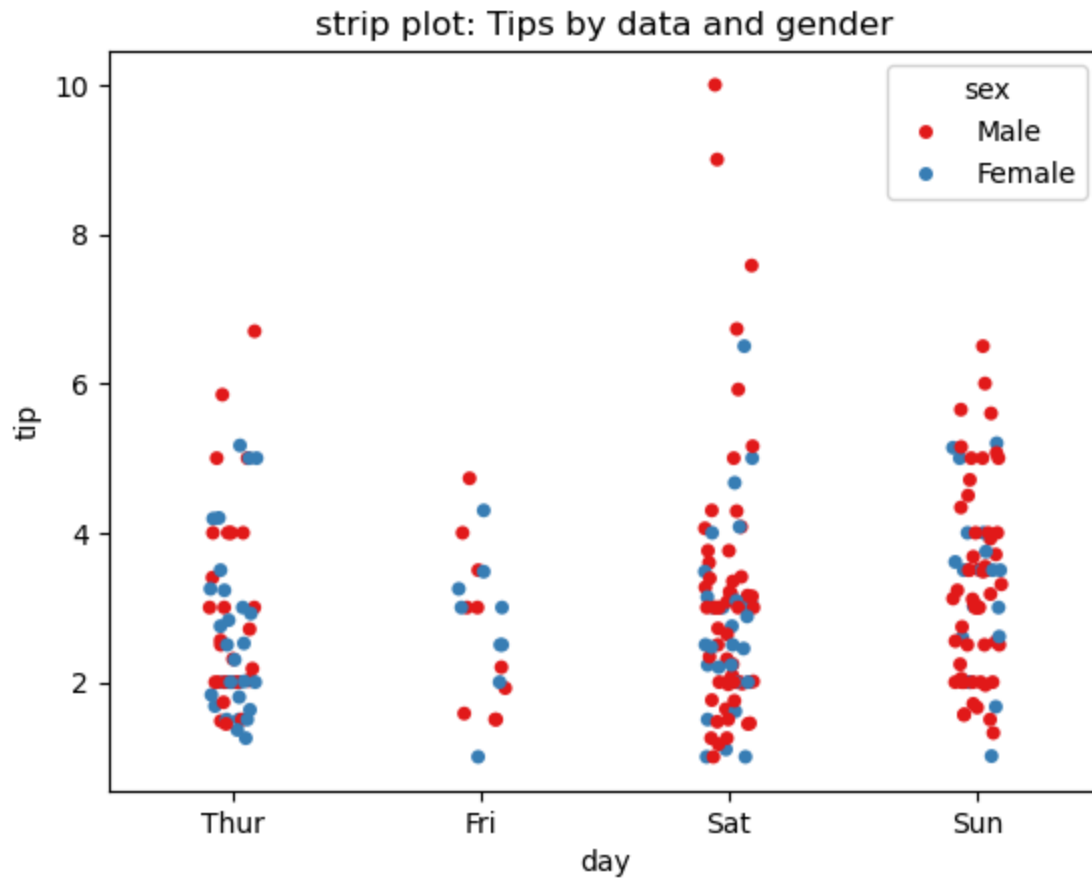
```
g = sns.FacetGrid(tips, col='time', row='smoker', margin_titles=True).map(sns.histp  
g
```

```
Out[23]: <seaborn.axisgrid.FacetGrid at 0x1fa79f91590>
```



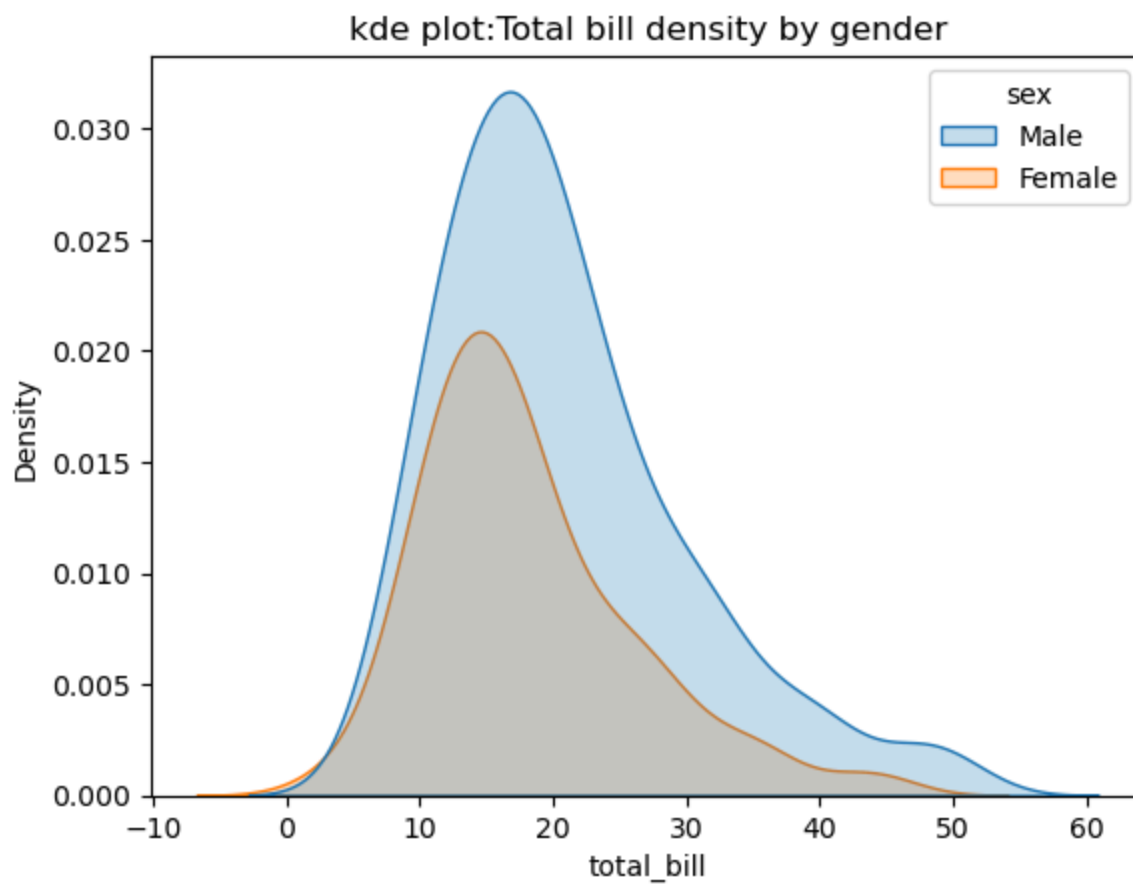
In [24]: #13. strip plot

```
sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='Set1')
plt.title("strip plot: Tips by data and gender")
plt.show()
```



In [25]: # 14. KDE PLOT

```
sns.kdeplot(data=tips, x='total_bill', hue='sex', fill=True, palette='tab10')  
plt.title("kde plot:Total bill density by gender")  
plt.show()
```

In []:

In []: